

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT2 — VISUALIZATION ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Analysis Report
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Evaluate the effectiveness of visualization design choices — proportional ink, axis scales, overplotting solutions, color use, and accessibility — and justify improvements.(BL5)</i>		
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Section / Batch:		Date:	27.08.2026

Code of Conduct

I Priyanka.G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: explain WHY the design choice is flawed, and justify your judgment.
- Where a question asks for a fix, propose a specific improvement — not just “make it better.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION ANALYSIS REPORT

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 46 Jittering & Transparency Categorical Scatterplot Overplotting (5 Marks)

Data: A categorical scatterplot of response times across 3 server clusters shows heavy point-stacking directly above each category label.

Task: Explain how combining jittering with partial transparency would improve this chart's readability.

Analysis Plan (Key Elements):

Principle Evaluated	Overplotting within a categorical axis position
Fix Technique 1	Jittering small random horizontal offset per point
Fix Technique 2	Partial transparency (alpha blending)
Combined Benefit	Recovers both point count and within-cluster density/shape

Critique & Recommendation:

When many points share, or sit very close to, the exact same categorical x-position, they stack directly on top of one another along a single vertical line per cluster. This makes two things impossible to read from the chart: how many observations actually belong to each cluster (since fully overlapping points render as one solid mark or a short solid line, no matter how many points are really there), and what the internal shape of each cluster's response-time distribution looks like (whether it is tightly concentrated, spread out, or has multiple sub-groups) all of that structure collapses into an undifferentiated blob.

Jittering adds a small random horizontal offset to each point's x-position, confined within its category's band, spreading the stack sideways so that points that would otherwise sit exactly on the same vertical line become visually separated from one another. This directly recovers the ability to see roughly how many points are in each cluster and gives a first impression of the distribution's spread, since points are no longer forced onto a single line.

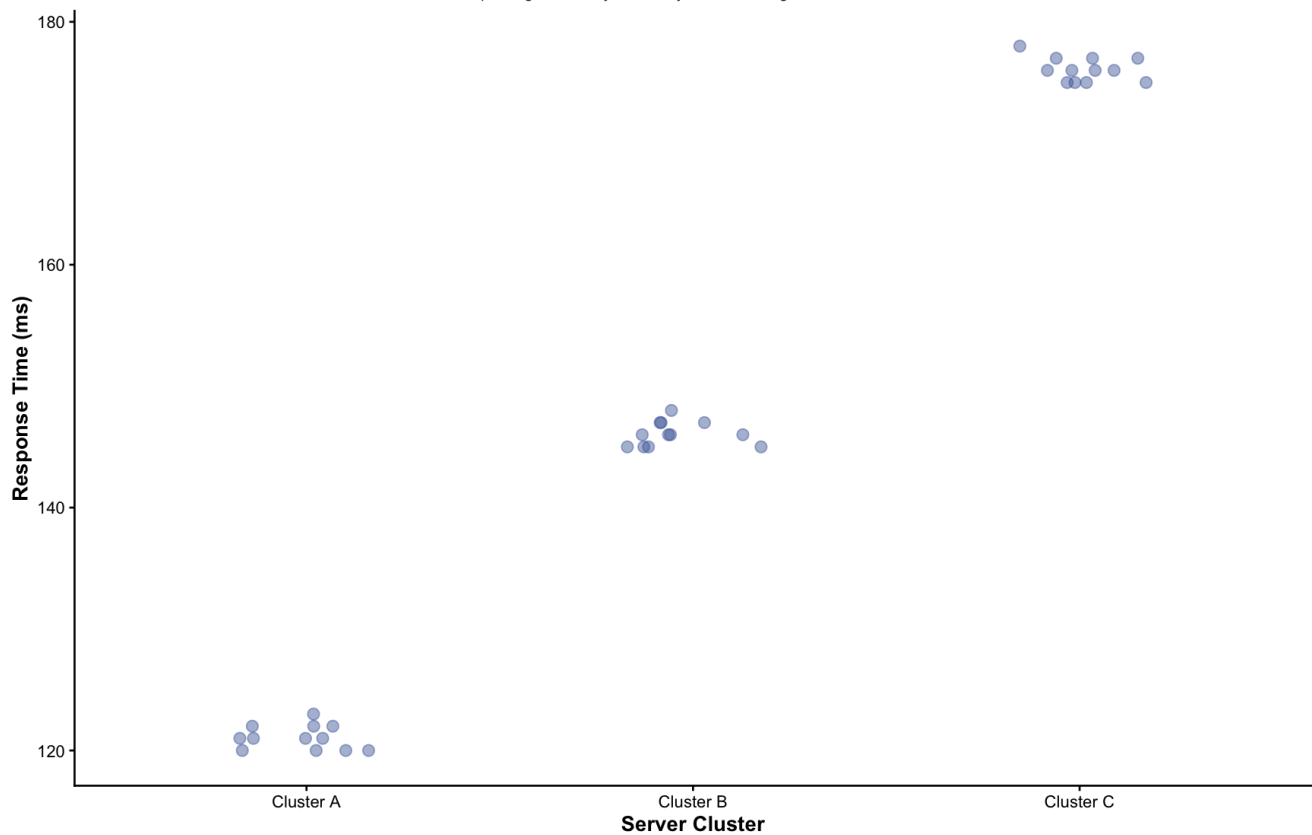
Partial transparency (alpha blending) complements jittering by handling the residual overlap jittering alone cannot fully resolve even after spreading points out, some will still land close together by chance. Giving each point a partial opacity (for example, 30–50%) means that regions where many points genuinely overlap will appear visibly darker or more saturated than regions with only a single point, reintroducing a density cue through visual darkness exactly where jittering's spatial spread is insufficient on its own.

Combined, the two techniques address two different aspects of the same overplotting problem: jittering solves the case where points exactly or near-exactly coincide by spatially separating them, while transparency solves the case where points still land close together after jittering by making that residual overlap visually informative (darker = more points) rather than visually indistinguishable. Together, a categorical scatterplot with moderate jitter (for example, a random offset within roughly $\pm 30\%$ of the category band width) and moderate opacity (around 0.3–0.5) lets a viewer see both the approximate number of observations per server cluster and the true shape and density of each cluster's response-time distribution both of which the original heavily-stacked plot concealed entirely.

Supporting Diagram:

192524124 Problem 46 - Jittering and Transparency

Improving readability of heavily stacked categorical observations



INTERPRETATION

Jittering separates overlapping points so individual observations become visible.
Partial transparency makes dense regions appear darker, revealing concentration.
Using both techniques reduces point-stacking and makes the distribution easier to read.

Figure 1: Categorical scatterplot of response times across the 3 server clusters

Problem 47 2D Histogram Age vs. Blood Pressure (5 Marks)

Data: A dataset of 5,200 (age, blood pressure) point pairs is too dense to read as a raw scatterplot.

Task: Explain what each cell's color would represent in a 2D histogram of this data, and what a very dark cell would indicate.

Analysis Plan (Key Elements):

Principle Evaluated	2D binning as a solution to severe overplotting
Cell Colour Encoding	Count (or density) of observations whose (age, BP) pair falls within that bin
Colour Scale	Sequential light for few/no points, dark for many points
Very Dark Cell	A strong local concentration a joint age–blood-pressure 'hotspot' in the data

Critique & Recommendation:

A 2D histogram divides the age–blood-pressure plane into a grid of rectangular bins, each spanning a fixed range of ages on one axis and a fixed range of blood-pressure values on the other. Each cell's colour encodes the count (or, equivalently, the density) of data points among the 5,200 (age, blood pressure) pairs whose combination of age and blood pressure falls within that specific cell's boundaries this is typically mapped through a sequential colour scale running from a light colour for cells containing few or no points up to a dark or highly saturated colour for cells containing many points.

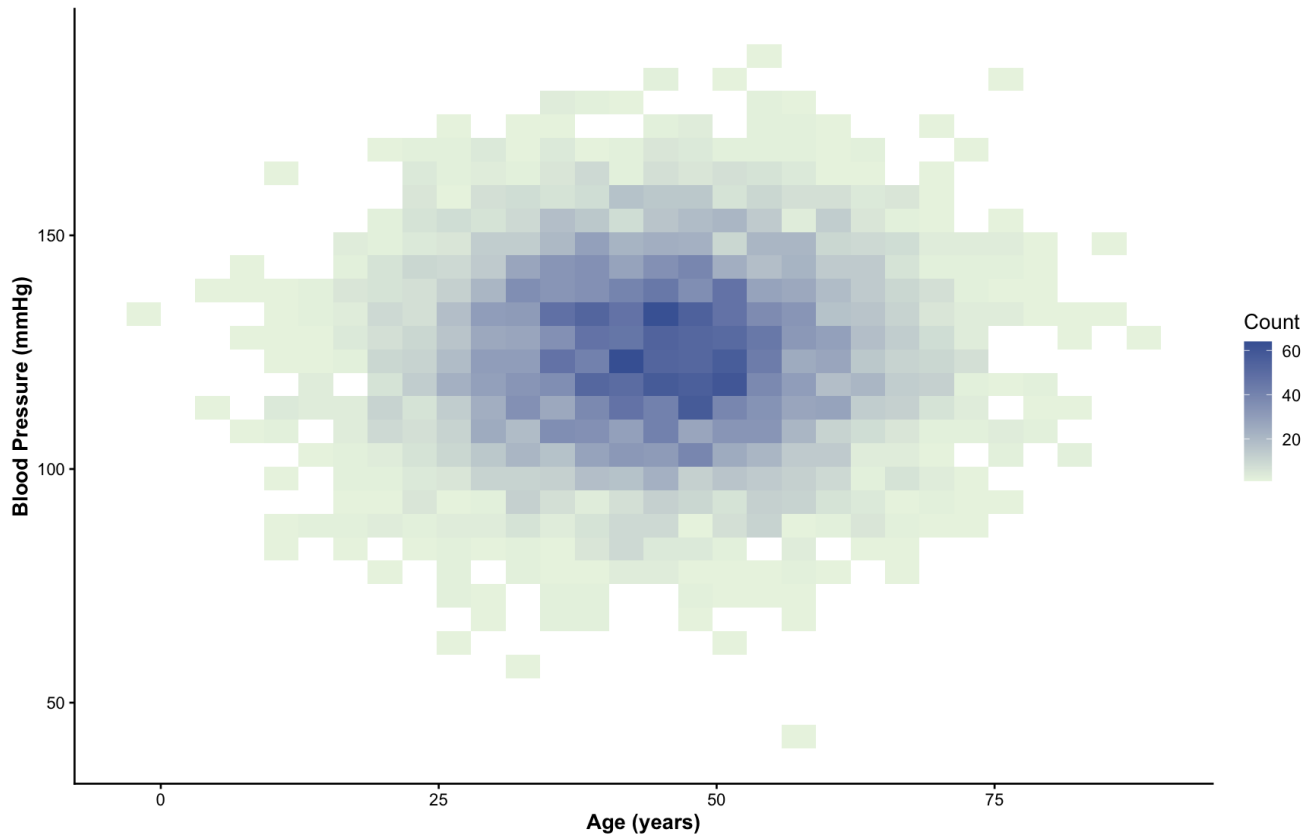
A very dark cell therefore indicates a bin containing a comparatively very high number of the 5,200 observations a strong local concentration of individuals who share both a similar age range and a similar blood-pressure range simultaneously. For example, a very dark cell centred around ages 50–55 with blood pressure 130–140 would indicate that a disproportionately large share of the sampled individuals in their early 50s have blood pressure readings specifically in that 130–140 range, revealing a mode, or 'hotspot,' in the joint age–blood-pressure distribution.

This directly solves the overplotting problem the raw scatterplot suffers from. With 5,200 individual points, a raw scatterplot would already render as a heavily saturated, largely black mass across much of its area, since far too many markers compete for too few distinguishable pixel positions exactly the overplotting failure mode seen with large point counts. The 2D histogram avoids this because the number of cells drawn is fixed by the chosen grid resolution rather than by the number of underlying records: cell darkness then legibly and precisely encodes exactly where the data is most concentrated, letting a reader immediately identify where the true joint hotspots in age and blood pressure lie, without needing to visually parse thousands of overlapping individual markers.

Supporting Diagram:

192524124 Problem 47 - 2D Histogram

Color intensity represents the concentration of observations



INTERPRETATION

Each cell represents a range of age and blood-pressure values.

The cell color represents the number of observations in that region.

A very dark cell indicates a high concentration of data points.

This approach is suitable for 5,200 observations because it reduces overplotting.

Figure 2: 2D histogram (binned grid) of the 5,200 age–blood-pressure pairs

Problem 48 Contour Line Spacing Rainfall vs. Crop Yield Density (5 Marks)

Data: A density plot of rainfall vs. crop yield is shown with contour lines overlaid.

Task: Explain what *tightly-packed contour lines* would indicate compared to *widely-spaced lines* in this context.

Analysis Plan (Key Elements):

Principle Evaluated	Contour lines connect points of equal density; their spacing encodes the gradient's steepness
Tightly-Packed Contours	A steep gradient density changes rapidly over a small region
Widely-Spaced Contours	A shallow gradient density changes slowly across a broader region
Agricultural Reading	Tight contours flag a narrow, sensitive rainfall–yield sweet spot

Critique & Recommendation:

Each contour line on a density plot connects points of equal density that is, every location along a single contour line represents the same concentration of underlying (rainfall, crop yield) observations, and consecutive contour lines typically represent evenly-spaced density increments moving toward (or away from) a peak.

Tightly-packed contour lines lines drawn very close together indicate a steep density gradient: the concentration of rainfall–yield data points is changing rapidly over a small region of the plot. This corresponds to a sharp, narrow peak or hotspot in the joint distribution, meaning a fairly specific rainfall range is associated with a tightly concentrated crop-yield outcome, and moving even a small distance away from that peak in either the rainfall or yield direction causes the observed density to drop off quickly.

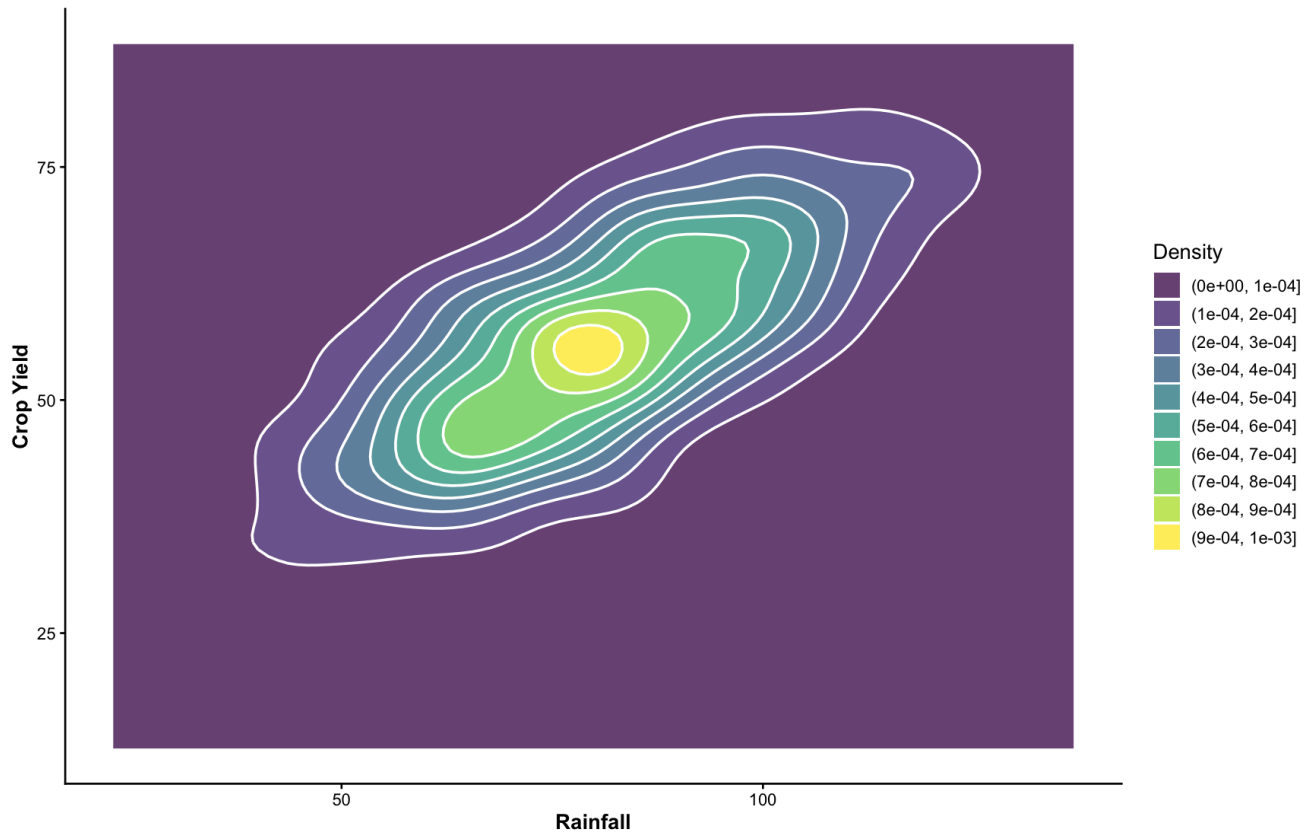
Widely-spaced contour lines, by contrast, indicate a shallow density gradient: the concentration changes slowly across that region of the plot, meaning the data is distributed relatively evenly or diffusely across a broader range of rainfall–yield combinations, forming a wide, gently-sloping plateau or broad hump rather than a tall, narrow spike.

In this agricultural context, tightly-packed contours around a specific rainfall level would suggest that crop yield responds very sensitively and precisely to rainfall within that narrow band useful information for identifying a tightly-defined optimal rainfall target for maximising yield. Widely-spaced contours elsewhere on the same plot would instead suggest that yield in that region is more tolerant of a broader range of rainfall amounts, i.e., less sensitive to the exact rainfall level a materially different, and equally useful, practical conclusion that the contour spacing alone communicates without needing to read any numeric density values directly.

Supporting Diagram:

192524124 Problem 48 - Rainfall and Crop Yield Density

Contour spacing shows how rapidly data density changes



INTERPRETATION

Tightly packed contour lines indicate a rapid change in data density over a small range of rainfall and crop-yield values.

Widely spaced contour lines indicate a more gradual change in density.

Therefore, contour spacing helps identify how sharply the distribution changes.

Figure 3: Density plot of rainfall vs. crop yield with contour lines overlaid, showing a region of tightly-packed contours

Problem 49 Arbitrary Colour Choice 7-Region Bar Chart (5 Marks)

Data: A bar chart shows 7 regions, each assigned an arbitrary color with no logic.

Task: Critique this color choice and propose a more purposeful, consistent alternative.

Analysis Plan (Key Elements):

Principle Evaluated	Colour should carry meaning; unused or arbitrary colour channels risk misleading readers
Flaw Type	Colour assigned with no systematic rule redundant at best, misleading at worst
Risk 1	Inconsistency across charts forces readers to re-learn the region– colour mapping each time
Risk 2	Coincidentally similar colours can visually imply a relationship between unrelated regions

Critique & Recommendation:

Assigning each of the 7 regions a colour with no logic wastes the colour channel's potential to carry meaning. Since the bars already sit at distinct positions along a categorical x-axis and their heights already encode the actual data value, colour is not needed to identify which bar is which but it could still be used purposefully (for example, to group regions, encode a secondary variable, or at minimum stay visually consistent and memorable across a report), and an arbitrary assignment throws that opportunity away entirely.

Beyond simply being a missed opportunity, arbitrary colour choice creates real risks for a reader. First, inconsistency across related charts: if the same 7 regions are coloured differently in another chart later in the same report, a reader has to re-learn the region-to-colour mapping each time, adding unnecessary cognitive load for no benefit. Second, accidental implied relationships: if two unrelated regions happen to receive similar colours purely by chance, a reader may visually group them as related or similar when no such relationship exists in the underlying data an unintended and misleading signal introduced entirely by the arbitrary colour assignment. Third, accessibility risk: an arbitrary palette has no guarantee of being colourblind-safe and could easily include colour pairs that are difficult or impossible for colourblind readers to distinguish, since no such consideration was applied when the colours were chosen.

The recommended alternative depends on whether the 7 regions have any real secondary structure worth encoding:

- If the regions have no meaningful secondary grouping or order, use one consistent neutral colour (e.g., a single shade of blue) for all 7 bars. Colour is not required to encode categorical identity when the x-axis labels already do that job, so a single consistent colour removes the arbitrary and potentially misleading signal entirely and lets bar height alone carry the comparison.
- If the regions do have a meaningful secondary structure (for example, grouped into larger zones such as North/South/East, or ranked by population size), use a colour scheme that reflects that real structure such as a categorical, colourblind-safe palette (e.g., ColorBrewer's Set2 or Dark2) with one distinct hue per zone or group so colour becomes a purposeful, additional encoding of a genuine variable rather than a decorative and potentially misleading one.

In either case, the same region-to-colour mapping should be kept identical across every chart in the same report, preserving reader familiarity and preventing the re-learning cost that inconsistent, arbitrary colour choices impose.

Supporting Diagram:

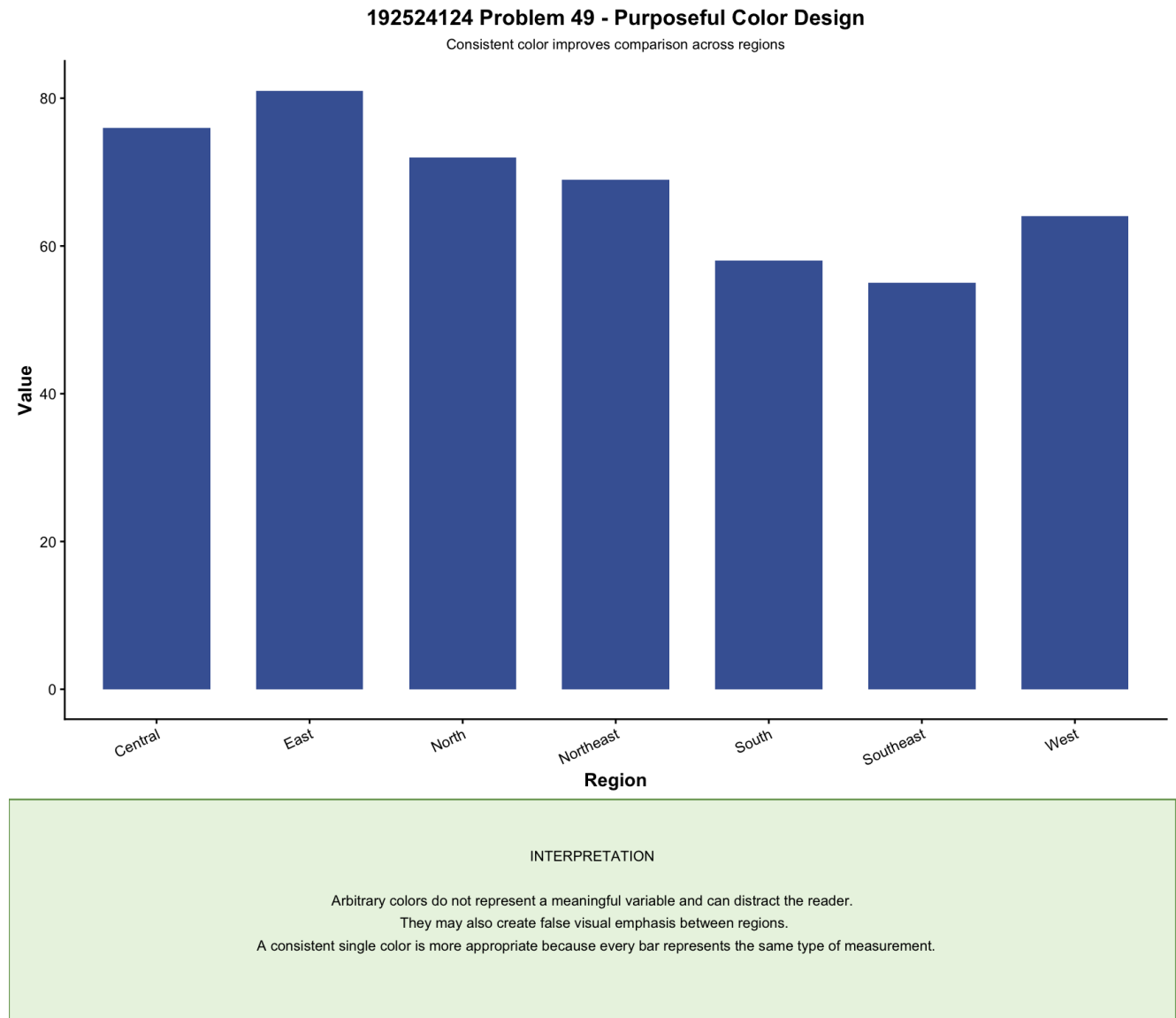


Figure 4: Bar chart of the 7 regions redrawn with the recommended colour scheme

Problem 50 Overloaded Encoding 15-Point Chart with Redundant Colour Scale (5 Marks)

Data: A chart displays 15 data points combining color, size, shape, and a redundant secondary color scale.

Task: Evaluate whether this design is effective for a general audience, and propose a simplification.

Analysis Plan (Key Elements):

Principle Evaluated	Number of simultaneous visual encodings should match the audience's decoding capacity
Encodings Used	Colour + size + shape + a redundant second colour scale four channels for 15 points
Flaw Type	Encoding overload: cognitive load far exceeds what the small dataset requires
Recommended Fix	Reduce to position + at most one channel, with direct text labels

Critique & Recommendation:

This design is not effective for a general audience. It layers four simultaneous visual encodings onto only 15 data points: colour (presumably encoding one variable), size (encoding a second), shape (encoding a third, likely categorical, variable), and then an additional redundant secondary colour scale layered on top of the colour channel already in use. With only 15 points already a small, easily-labelled dataset this adds a disproportionately high decoding burden relative to the actual amount of information being conveyed: a general-audience reader must simultaneously hold in mind what each shape means, what each size means, what the colour means, and then reconcile that with a second, redundant colour scale duplicating information already shown once, multiplying the effort required to correctly interpret even a single point before any cross-point comparison can even begin.

The redundant secondary colour scale is a particular red flag on its own. Encoding the same, or a highly correlated, variable through colour twice adds no new information to the chart while still consuming a design channel and visual space typically a second legend that could otherwise support genuine simplicity. Its presence signals that the design was not critically reviewed for redundancy before being finalised, since removing it would cost the chart nothing in terms of information conveyed.

With only 15 points, encodings should be reduced to the minimum needed to convey the intended story clearly ideally just two channels: position (the x and y axes, which the human eye reads more accurately than colour, size, or shape) plus at most one additional channel, chosen as whichever single variable is most important to the specific insight the chart needs to communicate. Each of the 15 points should also be directly labelled with a short text label rather than relying on shape or colour for identification, since with so few points direct labelling is entirely feasible and removes any need to decode a legend to know which point is which.

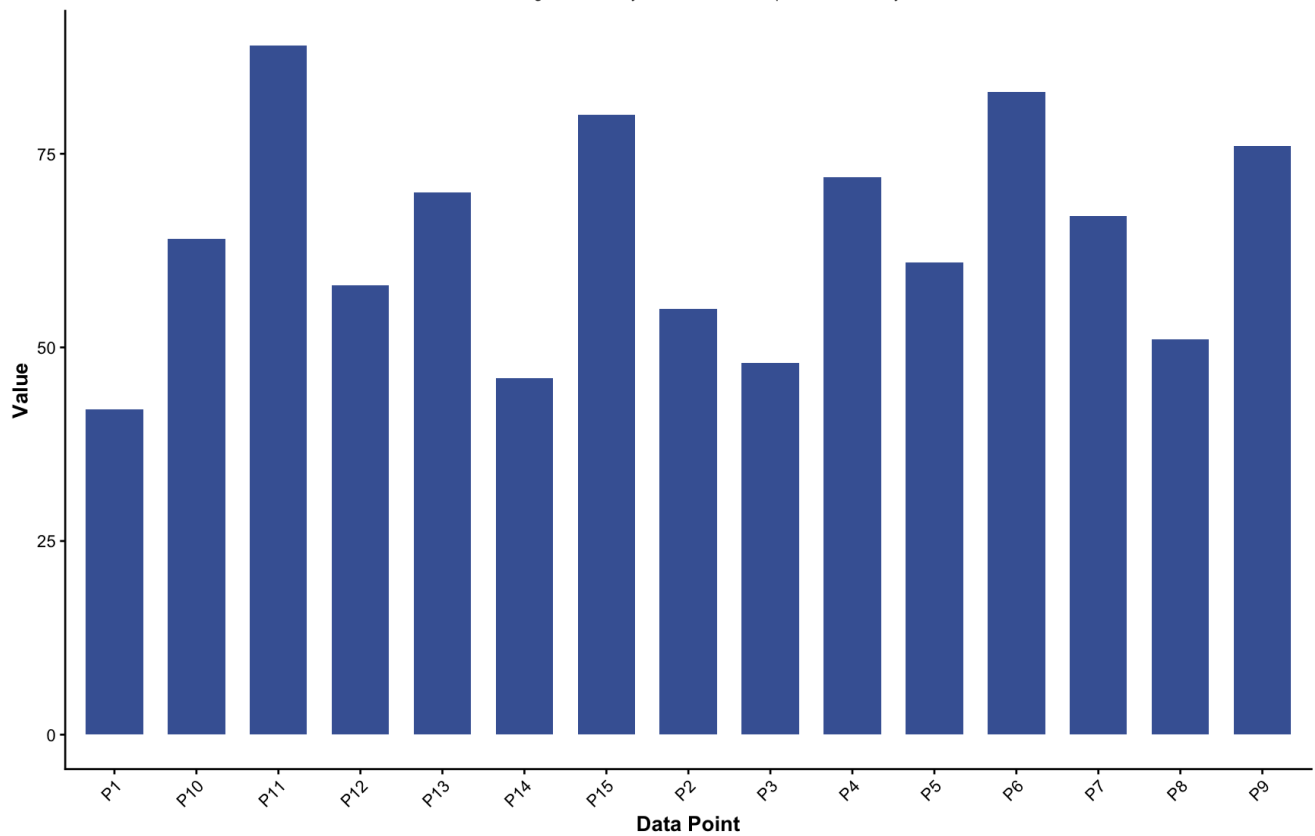
If two additional variables genuinely must be shown beyond position, the better approach is to split them into two small side-by-side charts (small multiples) rather than layering both onto a single, already-complex chart this keeps each individual chart simple and quickly

readable while still conveying all the intended information, just presented sequentially across two charts rather than crammed simultaneously into one.

Supporting Diagram:

192524124 Problem 50 - Simplifying Visual Encodings

Reducing unnecessary visual channels improves readability



INTERPRETATION

Using color, size, shape, and a redundant secondary color scale creates unnecessary complexity.
A general audience may find it difficult to identify the main message.
The visualization should be simplified by using one primary visual encoding, such as position or bar length.

Figure 5: Simplified redesign of the 15-point chart